A Curvature-Scale Coincidence on Compact S^3
and Two Dynamical Completion Tests

Technical preprint on a constant-positive-curvature R^{-2} scale

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Abstract. We study compact, simply connected spatial geometry $\Sigma \simeq S^3$ and the associated curvature-density scale on $S^3(R)$. The algebraic core is the identity $\rho_P \ell_P^2 = c^4/G$: in the tested integer family $\rho_\beta(n) = \rho_P \varphi^{-\beta n}$, the branch $\beta = 2$ is exactly $\rho_2(R) = c^4/(GR^2)$ and is the only branch among $\beta = 1, 2, 3, 4$ that reaches the observed dark-energy density at cosmological radii. The φ ladder is therefore used here as a logarithmic bookkeeping coordinate unless an independent integer-level selection principle is supplied. For constant positive curvature, and in particular on compact $S^3(R)$, the spatial Einstein–Hilbert scalar-curvature integral satisfies $\int_{S^3} \sqrt{g}\,{}^{(3)}R\, d^3x = 12\pi^2 R$, giving the natural positive scalar-curvature scale

$$\rho_{\mathrm{EH}}^{(3)}(R) = \frac{3}{8\pi}\frac{c^4}{GR^2}.$$

The coefficient $3/(8\pi)$ is a constant-positive-curvature density coefficient; compactness makes the total integral finite but does not make the local density coefficient topological. Evaluated at the observationally defined radius $R_{\mathrm{eff}} = R_H/\sqrt{\Omega_\Lambda} = c/(H_0\sqrt{\Omega_\Lambda})$, this reproduces $\rho_\Lambda = \Omega_\Lambda \rho_c$ exactly because R_{eff} is defined from the observed dark-energy scale. This equality is not a derivation of dark energy: Ω_Λ and R_{eff} are inserted from the Friedmann decomposition, and a conserved R^{-2} component has $w = -1/3$, not $w \simeq -1$. We therefore separate four layers: (i) the robust algebraic scaling result, (ii) the geometrically normalized positive scalar-curvature scale, (iii) the observationally inserted radius identification, and (iv) two incomplete dynamical completion tests–a CMC/York phase-drift ansatz and post-recombination null-domain kinematics, whose product-rule identity is $\ddot{D}_\gamma = \ddot{R}\chi_\gamma + c\dot{R}/R$. The resulting program is falsifiable through global curvature bounds, closed-geometry distances, low-ℓ CMB/topology tests, Pantheon+ supernovae, DESI DR2 BAO, and the sign and magnitude of the dark-energy equation-of-state evolution.

Keywords: compact S^3 topology; vacuum energy; curvature density; φ-bookkeeping; Einstein–Hilbert action; CMC phase law; null-domain kinematics; dark energy

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1 Introduction and Scope

The observed dark-energy density is of order $\rho_{\Lambda,\mathrm{obs}} \simeq 5.25 \times 10^{-10}\,\mathrm{J\,m^{-3}}$ for the Planck 2018 baseline values $H_0 \simeq 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_\Lambda \simeq 0.685$ [1]. Its smallness relative to Planck-scale vacuum estimates is the cosmological-constant problem [2]. The aim of this note is narrow: to test whether compact positive curvature and the associated R^{-2} scaling can reproduce the *scale* of $\rho_{\Lambda,\mathrm{obs}}$ without claiming a complete derivation of late-time acceleration.

The proposed geometry is a closed spatial section $\Sigma \simeq S^3$. The proposed scale ladder is

$$R_n = \ell_{\mathrm{P}}\varphi^n, \qquad \varphi = \frac{1+\sqrt{5}}{2}. \tag{1}$$

The main result is that the branch $\rho \propto R^{-2}$ is uniquely selected within the small integer family tested in Sec. 4. The same branch is generated by the spatial Einstein–Hilbert scalar-curvature integral on S^3. This establishes a curvature-scale coincidence, not a derivation of the late-time dark-energy component.

Status convention

The paper uses three labels. *Result* means an algebraic, numerical, or geometric statement derived in the paper. *Candidate* means a proposed completion requiring additional derivation or data fitting. *Open problem* means no claim of solution is made here.

Nonclaims. We do not prove that dark energy is topology, do not remove Λ from the gravitational action, do not infer cosmic topology from large-scale structure alone, and do not identify R_{eff} with the global FLRW curvature radius. The result is a geometrically normalized scale-setting observation plus a set of conservative dynamical tests.

2 Compact S^3 Geometry

Spatial topology

The spatial manifold is assumed to be compact, orientable, boundaryless and simply connected: $\Sigma \simeq S^3$. By the Poincare–Perelman theorem, a closed simply connected 3-manifold is homeomorphic to S^3 [3, 4]. Spherical quotient spaces S^3/Γ remain possible extensions, but the minimal case studied here is the simply connected one.

With curvature radius $R(t)$, the closed FLRW line element is

$$ds^2 = -c^2dt^2 + R^2(t)\left[d\chi^2 + \sin^2\chi\, d\Omega_2^2\right], \qquad 0 \le \chi \le \pi. \tag{2}$$

When comparison with FLRW observables is required, we write $R(t) = a(t)R_c$ with $a_0 = 1$, so that R_c is the present global curvature radius. The effective normalization radius R_{eff} introduced below is not identified with R_c.

The standard geometric quantities are

$$V_{S^3}(R) = 2\pi^2 R^3, \tag{3}$$

$${}^{(3)}R = \frac{6}{R^2}, \tag{4}$$

$$V_{\mathrm{ball}}(\chi, R) = 2\pi R^3\left(\chi - \frac{1}{2}\sin 2\chi\right), \tag{5}$$

$$A(\chi, R) = 4\pi R^2 \sin^2\chi. \tag{6}$$

The dimensionless volume-growth kernel of a geodesic ball is

$$K(\chi) = \frac{1}{V_{\text{ball}}}\frac{dV_{\text{ball}}}{d\chi} = \frac{2\sin^2\chi}{\chi - \sin\chi\cos\chi}. \tag{7}$$

It has $K(\chi) \sim 3/\chi$ as $\chi \to 0$ and $K(\chi) \to 0$ as $\chi \to \pi$.

3 The φ-Hierarchy and the Two-Index Correction

The geometric hierarchy is

$$R_n = \ell_{\text{P}}\varphi^n, \qquad V_n = 2\pi^2\ell_{\text{P}}^3\varphi^{3n}, \qquad A_n = 4\pi\ell_{\text{P}}^2\varphi^{2n}, \qquad n_R(R) = \frac{\ln(R/\ell_{\text{P}})}{\ln\varphi}. \tag{8}$$

φ-hierarchy status

Because $n_R = \ln(R/\ell_P)/\ln\varphi$ is allowed to be non-integer in the present paper, the golden-ratio hierarchy is currently a logarithmic coordinate and bookkeeping device. It does not by itself explain the vacuum scale. A physical φ-hierarchy would require an independent principle selecting integer levels, phase locking, spectral gaps, or a genuinely φ-recursive dynamical structure.

For the Planck 2018 baseline,

$$n_R(R_H) = 291.5436, \qquad n_R(R_{\text{eff}}) = 291.9367. \tag{9}$$

Thus a cosmic radius is a geometric level near 292, not near 85.

Level 85 is not a geometric cosmic radius

Under $R_n = \ell_{\text{P}}\varphi^n$, one finds

$$R_{85} = 9.3855 \times 10^{-18}\,\text{m}, \tag{10}$$

which is sub-nuclear. Any separate "85-level" used in auxiliary notes must therefore be interpreted as a distinct amplitude or calibration index, not as R_{85} in the geometric ladder.

A possible two-index relation is

$$n_A = n_R - n_{\text{cal}}, \qquad n_{\text{cal}} \simeq 207, \tag{11}$$

for which $R_{207} = 2.9441 \times 10^8\,\text{m}$ and $n_A(R_H) \simeq 84.54$. Equation (11) is only a calibration hypothesis.

4 Density Scaling Family and Selection of $\beta = 2$

Let $E_n = E_P\varphi^{\alpha n}$ and divide by $V_n \propto \varphi^{3n}$. Then

$$\rho_n = \rho_{\text{P}}\varphi^{-\beta n}, \qquad \beta = 3 - \alpha, \tag{12}$$

where the Planck energy density is

$$\rho_{\text{P}} = \frac{c^7}{\hbar G^2} = 4.6329 \times 10^{113}\,\text{J}\,\text{m}^{-3}. \tag{13}$$

The matching level and radius for a target density $\rho_{\Lambda,\text{obs}}$ are

$$n_\beta^* = \frac{\ln(\rho_{\text{P}}/\rho_{\Lambda,\text{obs}})}{\beta\ln\varphi}, \qquad R_\beta^* = \ell_{\text{P}}\varphi^{n_\beta^*}. \tag{14}$$

Table 1: β-branch matching table. The radius labels are corrected: $\beta = 3$ is Earth-scale, and $\beta = 4$ is sub-millimetre.

β	α	$\rho \propto$	n^*_β	R^*_β	Status
1	2	R^{-1}	588.2904	1.43×10^{88} m	excluded: super-cosmic
2	**1**	$\mathbf{R^{-2}}$	**294.1452**	$\mathbf{4.80 \times 10^{26}}$ m = **50.73 Gly**	**scale-compatible**
3	0	R^{-3}	196.0968	1.55×10^{6} m	excluded: planetary
4	-1	R^{-4}	147.0726	8.81×10^{-5} m	excluded: sub-mm

Because $\rho_{\rm P}\ell_{\rm P}^2 = c^4/G$, the $\beta = 2$ branch is exactly

$$\rho_2(R) = \frac{c^4}{GR^2}. \tag{15}$$

The coefficient in Eq. (15) is unity. The spatial Einstein–Hilbert normalization derived below supplies a geometric coefficient $C_{\rm EH} = 3/(8\pi)$, moving the equality radius from 50.73 Gly to 17.53 Gly.

Scaling-level result

Within the tested integer family $\beta = 1, 2, 3, 4$, only $\beta = 2$ places the observed vacuum-density scale at cosmological radii. This is a scaling result, not a derivation of the dark-energy equation of state.

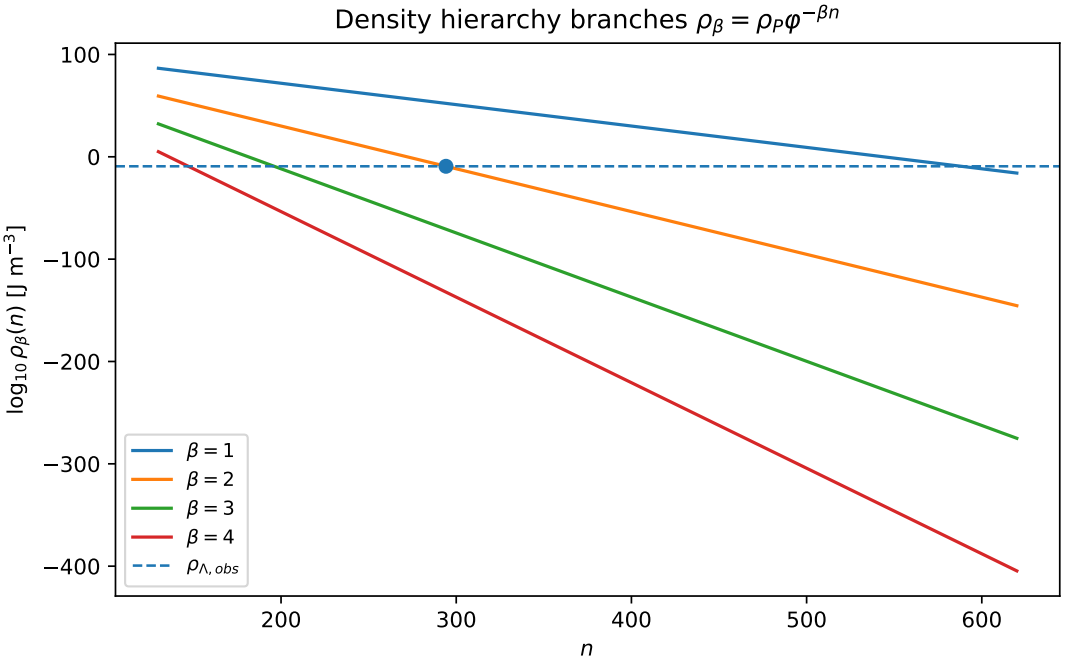


Figure 1: Density scaling $\rho_\beta(n) = \rho_{\rm P}\varphi^{-\beta n}$ for $\beta = 1, 2, 3, 4$. Only $\beta = 2$ intersects $\rho_{\Lambda,\rm obs}$ at a cosmological level.

5 Einstein–Hilbert Curvature Normalization on S^3

On $S^3(R)$,

$$\int_{S^3} \sqrt{g}\,{}^{(3)}\!R\, d^3x = \frac{6}{R^2}\, 2\pi^2 R^3 = 12\pi^2 R. \tag{16}$$

The integral scales linearly with R. Thus an EH-type curvature-energy scale is proportional to R, and the corresponding density is proportional to R^{-2}.

Define

$$E^{(3)}_{\mathrm{curv}}(R) = \frac{c^4}{16\pi G}\int_{S^3}\sqrt{g}\,{}^{(3)}R\,d^3x = \frac{3\pi c^4}{4G}R. \tag{17}$$

Dividing by $V_{S^3} = 2\pi^2R^3$ gives

$$\rho^{(3)}_{\mathrm{EH}}(R) = \frac{E^{(3)}_{\mathrm{curv}}}{V_{S^3}} = \frac{3}{8\pi}\frac{c^4}{GR^2}. \tag{18}$$

Geometric normalization

$$\boxed{C_{\mathrm{EH}} = \frac{3}{8\pi} = 0.1193662073.} \tag{19}$$

This is the constant-positive-curvature scalar-density coefficient. Compactness makes the total spatial integral finite; it does not by itself make the local density coefficient topological. The same density coefficient is obtained on spherical quotients S^3/Γ after division by their volume.

Hamiltonian status

Equation (18) is a spatial EH curvature-density scale. It is not a complete ADM Hamiltonian energy, since the full Hamiltonian constraint also contains extrinsic curvature, lapse dependence, matter, and possible subtraction or reference terms. The paper therefore uses $\rho^{(3)}_{\mathrm{EH}}$ as a scale-setting density, not as a closed gravitational energy theorem. In particular, no conserved quasi-local or global gravitational energy is being claimed for the compact slice.

Sign of the FLRW curvature fluid

The positive sign of $\rho^{(3)}_{\mathrm{EH}}$ is a convention for a scalar-curvature scale. It is not the sign of the effective FLRW curvature density. In the standard closed Friedmann equation,

$$H^2 = \frac{8\pi G}{3c^2}\rho - \frac{c^2}{a^2R_c^2}, \tag{20}$$

so moving the curvature term to the matter side gives

$$\rho_k = -\frac{3c^4}{8\pi G a^2 R_c^2} \quad (k=+1), \tag{21}$$

which is negative. Therefore $\rho^{(3)}_{\mathrm{EH}}$ must not be confused with the standard closed-curvature fluid.

6 Critical-Density Identity and the Tautology Warning

The Hubble radius is $R_H = c/H_0$. With energy density units,

$$\rho_{\mathrm{crit}} = \frac{3H_0^2c^2}{8\pi G} = \frac{3}{8\pi}\frac{c^4}{GR_H^2} = C_{\mathrm{EH}}\frac{c^4}{GR_H^2}. \tag{22}$$

The observed dark-energy density is

$$\rho_{\Lambda,\mathrm{obs}} = \Omega_\Lambda\rho_{\mathrm{crit}} = \frac{3\Omega_\Lambda}{8\pi}\frac{c^4}{GR_H^2} = C_{\mathrm{obs}}\frac{c^4}{GR_H^2}, \qquad \frac{C_{\mathrm{obs}}}{C_{\mathrm{EH}}} = \Omega_\Lambda. \tag{23}$$

Tautology warning

Substituting $R = R_H = c/H_0$ into $c^4/(GR^2)$ reproduces the critical density by definition. Likewise, $\rho^{(3)}_{\rm EH}(R_{\rm eff}) = \rho_\Lambda$ is exact only because $R_{\rm eff} = R_H/\sqrt{\Omega_\Lambda} = c/(H_0\sqrt{\Omega_\Lambda})$ is the observed dark-energy length written in this normalization. Equivalently, with $H_\Lambda = H_0\sqrt{\Omega_\Lambda}$, one has $R_{\rm eff} = c/H_\Lambda$. The construction therefore reproduces ρ_Λ because $R_{\rm eff}$ is defined from ρ_Λ. The nontrivial task is to derive Ω_Λ, $R_{\rm eff}$, or a dynamical selection rule from compact geometry rather than inserting them from data.

Define the effective curvature-energy radius by

$$\rho^{(3)}_{\rm EH}(R_{\rm eff}) = \rho_{\Lambda,\rm obs}. \tag{24}$$

Using Eq. (23),

$$\boxed{R_{\rm eff} = \frac{R_H}{\sqrt{\Omega_\Lambda}} = 1.6583 \times 10^{26}\,\mathrm{m} = 17.53\,\mathrm{Gly}.} \tag{25}$$

This equality is exact given the definitions and the observed input Ω_Λ; it is the observed dark-energy length written in the present normalization. It is not a first-principles derivation of Ω_Λ.

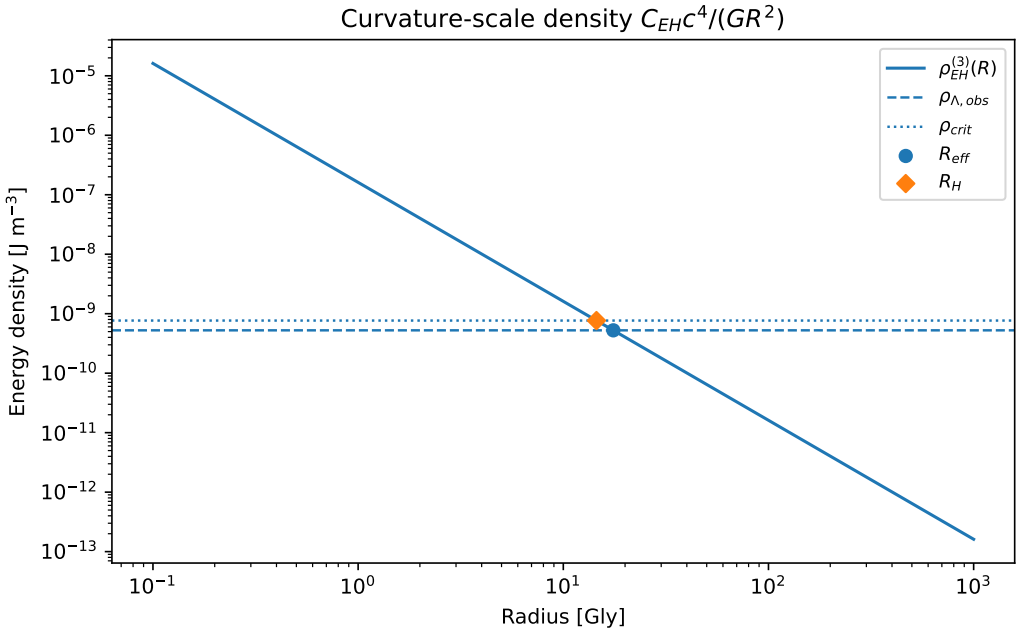


Figure 2: $\rho^{(3)}_{\rm EH}(R) = C_{\rm EH}c^4/(GR^2)$. The equality with $\rho_{\Lambda,\rm obs}$ occurs at $R_{\rm eff} = R_H/\sqrt{\Omega_\Lambda}$; the equality with $\rho_{\rm crit}$ at R_H is the critical-density identity.

7 Three Radii That Must Not Be Conflated

The construction contains three different radii.

Hubble radius.

$$R_H = \frac{c}{H_0} = 1.3725 \times 10^{26}\,\mathrm{m} = 14.51\,\mathrm{Gly}. \tag{26}$$

Effective curvature-energy radius. This is $R_{\rm eff} = R_H/\sqrt{\Omega_\Lambda} = 17.53\,\mathrm{Gly}$. If the Hubble radius is represented as an arc length on an auxiliary $S^3(R_{\rm eff})$, the angle is

$$\chi_H = \frac{R_H}{R_{\rm eff}} = \sqrt{\Omega_\Lambda} = 0.827647. \tag{27}$$

The corresponding geodesic-ball volume fraction is

$$f_{\rm obs}(\chi_H) = \frac{V_{\rm ball}(\chi_H, R_{\rm eff})}{V_{S^3}(R_{\rm eff})} = \frac{\chi_H - \frac{1}{2}\sin 2\chi_H}{\pi} = 0.104861. \tag{28}$$

This fraction is a diagnostic number, not yet a derived Ω_Λ.

Global FLRW curvature radius. For a closed FLRW model with present scale factor $a_0 = 1$,

$$R_{\rm curv} = \frac{R_H}{\sqrt{|\Omega_k|}}. \tag{29}$$

Planck combined with BAO is consistent with spatial flatness, with constraints of order $|\Omega_k| \lesssim 10^{-3}$–$10^{-2}$ depending on data and model choices [1]. Thus the global S^3 radius must be much larger than R_H.

Curvature conflict if $R_{\rm eff}$ is misidentified

If $R_{\rm eff}$ were identified with the global FLRW curvature radius, then

$$\Omega_k = -\left(\frac{R_H}{R_{\rm eff}}\right)^2 = -\Omega_\Lambda \simeq -0.685, \tag{30}$$

which is excluded. Therefore $R_{\rm eff}$ can only be an effective curvature-energy scale, not the global FLRW curvature radius.

Table 2: Physically distinct radii and φ levels.

Description	R [m]	R [Gly]	n_R	$n_R - 207$	Role
$R_H = c/H_0$	1.3725×10^{26}	14.51	291.5436	84.5436	observational
$R_{\rm eff} = R_H/\sqrt{\Omega_\Lambda}$	1.6583×10^{26}	17.53	291.9367	84.9367	effective scale
$R_{\rm curv}(\lvert\Omega_k\rvert = 10^{-2})$	1.3725×10^{27}	145.1	296.3285	89.3285	global lower bound
$R_{\rm curv}(\lvert\Omega_k\rvert = 10^{-3})$	4.3402×10^{27}	458.8	298.7210	91.7210	global lower bound
$R_{\rm curv}(\lvert\Omega_k\rvert = 10^{-4})$	1.3725×10^{28}	1450.7	301.1135	94.1135	global lower bound

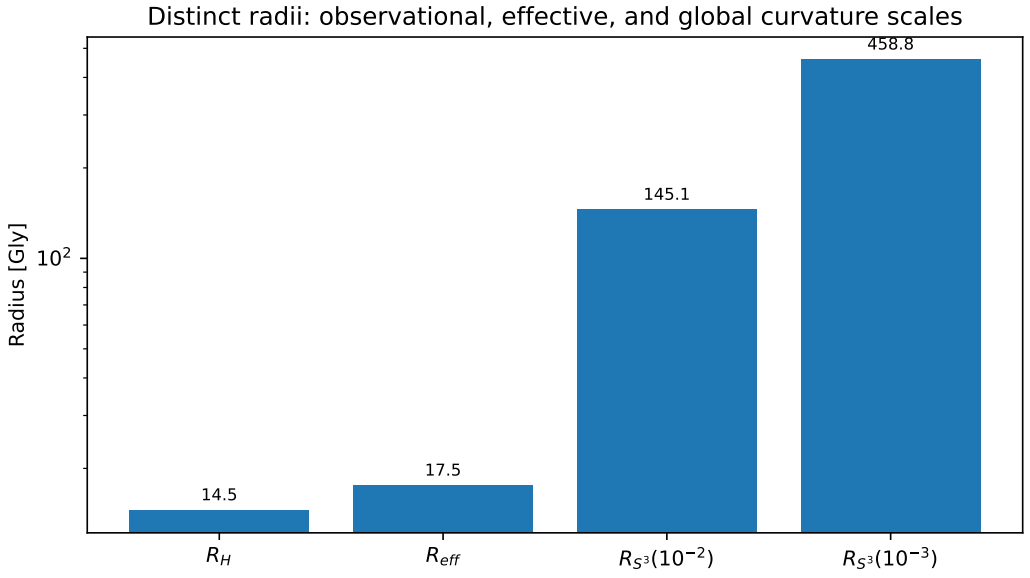


Figure 3: The Hubble radius, the effective curvature-energy radius, and two illustrative global curvature radii.

8 The Equation-of-State Gap

If $\rho_{\text{eff}}(a) = \rho_0 a^{-2}$ is treated as a conserved perfect-fluid energy density, the continuity equation gives

$$\rho(a) \propto a^{-3(1+w)}, \qquad -3(1+w) = -2, \qquad w = -\frac{1}{3}. \tag{31}$$

The acceleration equation in energy-density units is

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2}\rho(1+3w). \tag{32}$$

For $w = -1/3$ this gives no acceleration from that component alone.

Equation-of-state gap

The branch $\rho \sim R^{-2}$ gives the correct magnitude near cosmic radii, but as a conserved fluid it has $w = -1/3$, not $w \simeq -1$. A second dynamical principle is required if the branch is to model late-time acceleration.

The conservative interpretation is therefore not "dark energy is simply a^{-2} curvature fluid." Instead, the present paper uses $\rho^{(3)}_{\text{EH}}(R_{\text{eff}})$ as a normalization scale and asks whether a separate dynamical factor can keep the effective component nearly constant.

9 Candidate Test I: Post-Recombination Null-Domain Kinematics

After recombination, photons propagate freely and define an observable null domain. A radial null geodesic of Eq. (2) satisfies

$$\dot{\chi}_\gamma = \frac{c}{R(t)}. \tag{33}$$

Define a light-front arc length

$$D_\gamma(t) = R(t)\chi_\gamma(t). \tag{34}$$

The product rule gives

$$\dot{D}_\gamma = \dot{R}\chi_\gamma + R\dot{\chi}_\gamma = \dot{R}\chi_\gamma + c, \tag{35}$$

and

$$\ddot{D}_\gamma = \ddot{R}\chi_\gamma + 2\dot{R}\dot{\chi}_\gamma + R\ddot{\chi}_\gamma. \tag{36}$$

Since $\dot{\chi}_\gamma = c/R$,

$$\ddot{\chi}_\gamma = -\frac{c\dot{R}}{R^2}, \tag{37}$$

so

$$\boxed{\ddot{D}_\gamma = \ddot{R}\chi_\gamma + \frac{c\dot{R}}{R}.} \tag{38}$$

For a coasting scale $\ddot{R} = 0$ with $\dot{R} > 0$, this coordinate light-front length has $\ddot{D}_\gamma = c\dot{R}/R > 0$.

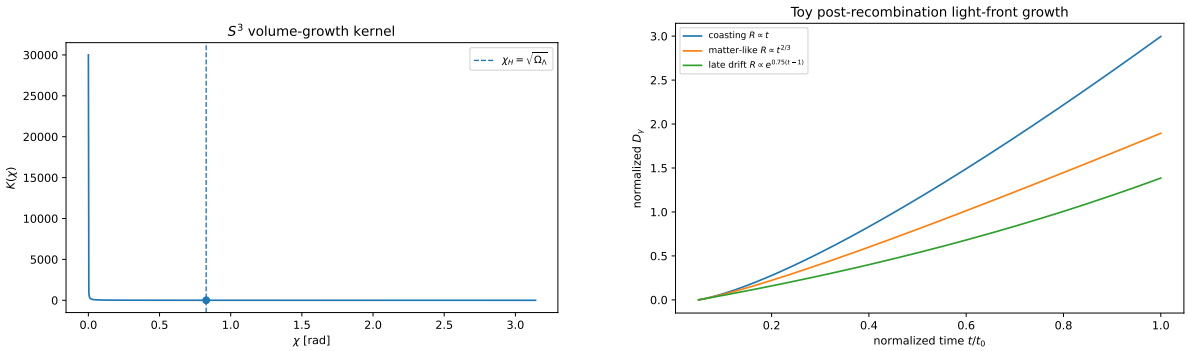
Interpretation limit

Equation (38) is a kinematic identity for the chosen physical arc-length variable, and by itself it does not distinguish the model from a standard FLRW description. Until D_γ is mapped into D_C, D_M, D_L, D_A and fitted against redshift observables, it is a coordinate-domain identity rather than a falsifiability test.

The observable null-domain volume is

$$V_\gamma(t) = 2\pi R^3(t)\left[\chi_\gamma(t) - \frac{1}{2}\sin 2\chi_\gamma(t)\right], \qquad \frac{1}{R}K(\chi_\gamma) = \frac{1}{V_\gamma}\frac{\partial V_\gamma}{\partial \chi_\gamma}. \tag{39}$$

Figure 4 shows the kernel and toy light-front behavior.



(a) S^3 volume-growth kernel. (b) Toy light-front growth.

Figure 4: Null-domain diagnostics. The plotted light-front curves are illustrative only and are not fits to data.

10 Candidate Test II: CMC/York Phase Drift

The second completion is a phase-clock ansatz in a compact CMC/York treatment. York-time quantum cosmology has been studied in related canonical contexts [5]. The present paper uses only the following ansatz as a candidate completion:

$$H(t) = \frac{1}{3}\dot{\Theta}(t) + \mathcal{O}(\epsilon_{\mathrm{BO}}^2), \tag{40}$$

where Θ is a gauge-invariant determinant-line geometric phase for a transported transverse-traceless sector. Equation (40) is not assumed to be established by existing quantum gravity; it is a candidate closure condition requiring independent derivation.

If $\dot{\Theta} \to \Theta_1$ at late times, then $H \to H_\infty = \Theta_1/3$ and

$$\Lambda_{\mathrm{eff}} = \frac{3H_\infty^2}{c^2}. \tag{41}$$

For an asymptotic phase-dominated regime, or for the total effective fluid after the usual Friedmann reduction, one may write

$$w_{\mathrm{eff}}(a) = -1 - \frac{2}{3}\frac{\dot{H}}{H^2}. \tag{42}$$

In a mixed matter-plus-dark-energy universe this is not automatically the separate dark-energy equation of state unless matter, radiation, and curvature have first been subtracted. If the time derivative in Eq. (40) is the same cosmic-time derivative, then

$$\dot{H} = \frac{1}{3}\ddot{\Theta}, \qquad H^2 = \frac{1}{9}\dot{\Theta}^2, \tag{43}$$

and hence

$$\boxed{w_{\mathrm{eff}} = -1 - 2\frac{\ddot{\Theta}}{\dot{\Theta}^2}.} \tag{44}$$

Slow phase-rate drift gives $w_{\mathrm{eff}} \simeq -1$.

Phase-law status

The phase law is an ansatz, not a result derived in this paper. A mathematical completion must define the family of Hilbert spaces $\mathcal{H}_t$, the connection ∇_t, the phase $\Theta(t)$, the TT-sector state or density matrix ρ_{TT}, the regularization, and the time parameter. The York-time literature supports the use of York time in canonical cosmology, but does not by itself establish the determinant-line phase law used here. Earlier sign problems in negative effective occupation numbers are therefore not repaired here; they define the next calculation.

11 Conservative Hybrid Parameterization

The conservative model separates scale from evolution:

$$\rho_{\mathrm{DE,eff}}(a) = \rho_{\mathrm{scale}} F_{\mathrm{dyn}}(a), \qquad \rho_{\mathrm{scale}} = \rho^{(3)}_{\mathrm{EH}}(R_{\mathrm{eff}}) = \rho_{\Lambda,\mathrm{obs}}, \qquad F_{\mathrm{dyn}}(1) = 1. \tag{45}$$

A minimal phenomenological proxy is

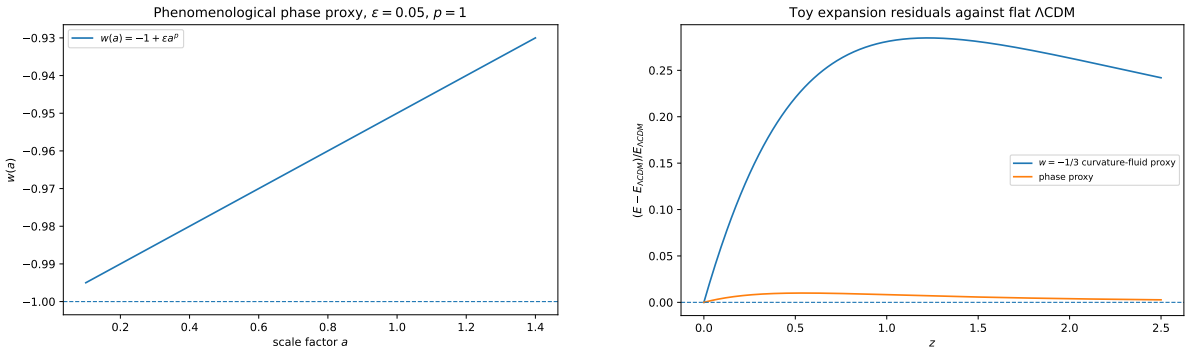
$$w(a) = -1 + \varepsilon a^p, \qquad F_{\mathrm{dyn}}(a) = \exp\left[-\frac{3\varepsilon}{p}(a^p - 1)\right], \qquad \varepsilon > 0, \quad p > 0. \tag{46}$$

At $a = 1$, $w_0 = -1 + \varepsilon$ and $w_a = -\varepsilon p$ in the usual $w(a) = w_0 + w_a(1 - a)$ convention. The minus sign in F_{dyn} follows from $d\rho/d\ln a = -3(1 + w)\rho$; for $\varepsilon > 0$ and $p > 0$ the component is quintessence-like, with larger density in the past than at $a = 1$.

The closed-geometry Friedmann equation should use a separate global curvature radius R_c:

$$H^2(a) = \frac{8\pi G}{3c^2}\left[\rho_{m0}a^{-3} + \rho_{r0}a^{-4} + \rho_{\Lambda,\mathrm{obs}}F_{\mathrm{dyn}}(a)\right] - \frac{kc^2}{a^2R_c^2}, \qquad k = +1. \tag{47}$$

The normalization radius R_{eff} in Eq. (45) is not R_c.



(a) Toy $w(a)$ proxy.

(b) Expansion residuals.

Figure 5: Phenomenological diagnostics. These panels illustrate the difference between an excluded pure $w = -1/3$ curvature-fluid interpretation and a near-Λ phase proxy. They are not likelihood fits.

12 Numerical Tables

13 Observational Protocol

For the global closed geometry, use the standard relations

$$D_C(z) = c\int_0^z \frac{dz'}{H(z')}, \tag{48}$$

$$D_M(z) = R_c \sin\left(\frac{D_C(z)}{R_c}\right), \tag{49}$$

$$D_L(z) = (1 + z)D_M(z), \qquad D_A(z) = \frac{D_M(z)}{1 + z}. \tag{50}$$

The model must be compared to supernova, BAO, and CMB data using these distances, not by fitting D_γ directly. Pantheon+ provides 1701 light curves of 1550 distinct Type Ia supernovae across $z = 0.001$–2.26 [7]. DESI DR2 BAO uses more than 14 million galaxies and quasars and,

Table 3: Constants and derived targets used in the numerical checks.

Quantity	Symbol	Value	Units/status
Speed of light	c	$2.9979245800 \times 10^{8}$	exact SI
Reduced Planck constant	$\hbar$	$1.0545718170 \times 10^{-34}$	J s
Newton constant	G	$6.6743000000 \times 10^{-11}$	SI
Planck length	ℓ_{P}	$1.6162550239 \times 10^{-35}$	m
Planck energy density	ρ_{P}	$4.6329467907 \times 10^{113}$	J m^{-3}
Golden ratio	φ	1.6180339887	exact
Hubble constant	H_0	$2.1842852411 \times 10^{-18}$	s^{-1}
Dark-energy fraction	Ω_Λ	0.6850000000	baseline input
Hubble radius	R_H	$1.3724968349 \times 10^{26}$	m
Critical density	ρ_{crit}	$7.6689477678 \times 10^{-10}$	J m^{-3}
Observed DE density	$\rho_{\Lambda,\mathrm{obs}}$	$5.2532292210 \times 10^{-10}$	J m^{-3}
$C_{\mathrm{EH}} = 3/(8\pi)$	C_{EH}	0.1193662073	derived
Effective radius	R_{eff}	$1.6583113220 \times 10^{26}$	m
Observable angle	χ_H	0.8276472679	rad
Ball fraction	f_{obs}	0.1048612055	diagnostic

Table 4: Coefficient candidates in $\rho = Cc^4/(GR^2)$.

C expression	C	$R(C)$ [Gly]	$R(C)/R_H$	Status
$3/(8\pi)$	0.119366	17.53	1.208	spatial EH coefficient
$3\Omega_\Lambda/(8\pi)$	0.081766	14.51	1.000	observational coefficient
$1/(2\pi^2)$	0.050661	11.42	0.787	geometric candidate only
$3/(4\pi^2)$	0.075991	13.99	0.964	geometric candidate only
1	1.000000	50.73	3.497	raw $\beta = 2$ Planck scaling

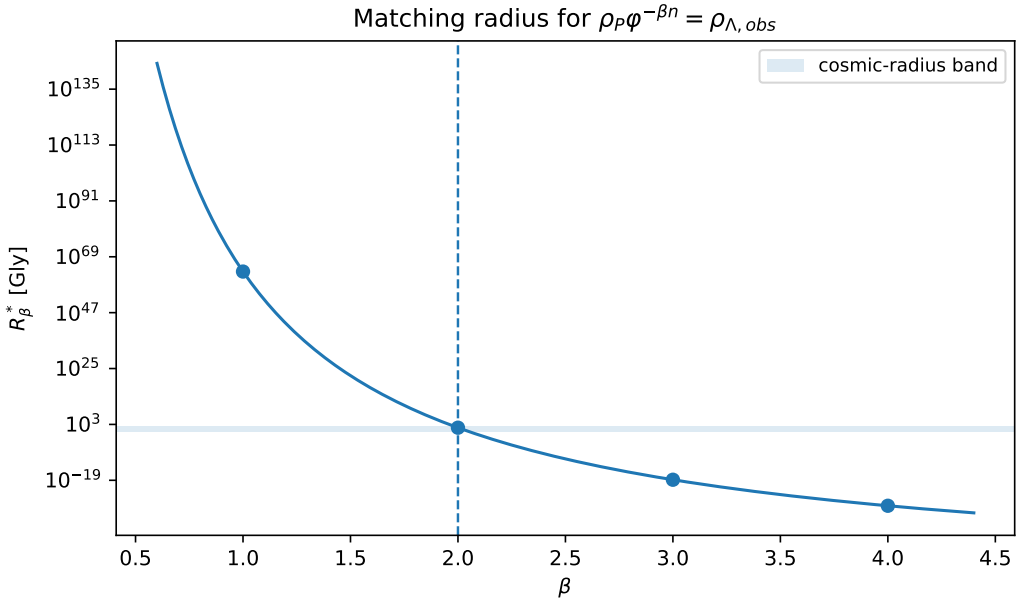


Figure 6: Matching radius as a function of β. The integer branch $\beta = 2$ lands in the cosmological-radius band; neighboring integer branches do not.

in combined $w_0 w_a$ fits, reports a favored quadrant $w_0 > -1$, $w_a < 0$ while still finding that BAO distances are well described by flat ΛCDM [6].

Falsifiability tests

1. **Equation-of-state sign and size.** The proxy in Eq. (46) with $\varepsilon > 0$ and $p > 0$ predicts $w_0 > -1$ and $w_a < 0$. Strong evidence for the opposite sign would falsify that realization.
2. **Pure curvature-fluid rejection.** If the model is forced to evolve as $\rho \propto a^{-2}$, it predicts $w = -1/3$ and must fail late-time acceleration fits.
3. **Curvature bound.** The global R_c must satisfy CMB+BAO curvature bounds. $R_c = R_{\text{eff}}$ is excluded.
4. **Topology tests.** Matched circles apply only to multiply connected quotients S^3/Γ, not to simply connected S^3. Non-detection constrains quotient models, not the minimal simply connected model by itself [8].
5. **Low-mode spectrum.** Compact positive curvature gives a discrete Laplacian spectrum. Any claimed low-ℓ CMB signature must be computed with proper transfer functions and likelihoods.

14 Internal Consistency Conditions

1. **Volume normalization.** Use $V_{S^3} = 2\pi^2 R^3$, not the Euclidean ball volume $4\pi R^3/3$, except in the local $\chi \ll 1$ limit.
2. **Coefficient separation.** $C_{\text{EH}} = 3/(8\pi)$ is geometrically derived. $C_{\text{obs}} = 3\Omega_\Lambda/(8\pi)$ is observational until Ω_Λ is derived.
3. **Branch labels.** The raw $\beta = 2$ Planck scaling has coefficient $C = 1$ and equality radius 50.73 Gly; the EH-normalized equality radius is 17.53 Gly.
4. **Radius separation.** R_{eff} must not be identified with the global FLRW curvature radius R_c.
5. **85-level disambiguation.** $n = 85$ is not a geometric cosmic level. It can only be an amplitude/calibration index if Eq. (11) is justified.
6. **Dimensional hygiene.** Expressions such as $3R_n + H$ are invalid. Use c/R_n, HR_n/c, or $3/R_n$ depending on the intended dimension.
7. **Matched circles.** They are not a direct test of simply connected S^3; they test multiply connected quotient identifications.
8. **CMC phase law.** It is a candidate closure, not a completed derivation.

15 Critical Weak Points

P1: Missing derivation of Ω_Λ

The ratio $C_{\text{obs}}/C_{\text{EH}} = \Omega_\Lambda$ is inserted from observation. A first-principles compact-S^3 derivation is still absent.

P2: Effective radius vs. global curvature radius

The model survives curvature bounds only if R_{eff} is an effective curvature-energy scale, not the global FLRW curvature radius. This separation needs an action-level derivation.

P3: No action for $F_{\rm dyn}$

The hybrid factor $F_{\rm dyn}(a)$ is phenomenological until it is derived from a Lagrangian, Hamiltonian constraint, or controlled semiclassical approximation.

P4: Null-domain distances not fitted

The identity for D_γ is a coordinate-domain identity until it is mapped to D_C, D_M, D_L, D_A, BAO distances, and redshift. It is not yet a falsifiability test by itself.

P5: CMC phase computation incomplete

A viable phase completion must specify $(\mathcal{H}_t, \nabla_t, \Theta(t), \rho_{\rm TT})$, the regularization and time parameter, then compute $\Theta(t)$ from a concrete compact quantum state while verifying positivity, gauge invariance, and time-convention stability.

16 Final Verdict

Four-Level Verdict

Level 1 – established algebraic scaling:

$$\rho_P \ell_P^2 = \frac{c^4}{G}, \qquad \beta = 2 \iff \rho_2(R) = \frac{c^4}{GR^2}.$$

Among the tested integer branches $\beta = 1, 2, 3, 4$, this is the only one that reaches the observed dark-energy density at cosmological radii.

Level 2 – established scalar-curvature normalization:

$$\int_{S^3} \sqrt{g}\,{}^{(3)}\!R\, d^3x = 12\pi^2 R \implies \rho_{\rm EH}^{(3)}(R) = \frac{3}{8\pi}\frac{c^4}{GR^2}.$$

This is a positive constant-curvature scalar-density scale, not the negative effective curvature fluid of the closed Friedmann equation, and not a uniquely topological density coefficient.

Level 3 – observationally inserted coincidence:

$$\rho_{\rm EH}^{(3)}(R_{\rm eff}) = \rho_\Lambda, \qquad R_{\rm eff} = \frac{R_H}{\sqrt{\Omega_\Lambda}} = 17.53\,\text{Gly}.$$

This identity is exact after inserting Ω_Λ: $R_{\rm eff} = c/(H_0\sqrt{\Omega_\Lambda}) = c/H_\Lambda$ is the observed dark-energy length in this normalization. Thus the equality is a curvature-scale coincidence, not a derivation from compact geometry.

Level 4 – open dynamics: A conserved R^{-2} component has $w = -1/3$ and does not drive late-time acceleration. The remaining problem is dynamical: derive $F_{\rm dyn}(a)$ from an action, phase law, or null-domain likelihood construction, or reject the model.

The paper establishes a robust curvature-scale coincidence: the unique tested integer density branch $\beta = 2$ gives $\rho \propto R^{-2}$, and the constant-positive-curvature scalar-density calculation fixes the natural coefficient $3/(8\pi)$. Compactness makes the total integral finite, but the local density coefficient is not uniquely topological. Evaluated at the observationally defined radius $R_H/\sqrt{\Omega_\Lambda} = c/H_\Lambda$, this reproduces ρ_Λ because that radius is defined from the observed dark-energy scale. This is not a derivation of dark energy, because Ω_Λ and the effective radius are inserted from the Friedmann

decomposition, and a conserved R^{-2} component has $w = -1/3$. The remaining problem is dynamical: derive $F_{\rm dyn}(a)$, or reject the model.

17 Next Calculations Required

1. **ADM closure.** Derive or reject Eq. (18) inside the full ADM Hamiltonian constraint with lapse, extrinsic curvature, and reference terms.
2. **Ω_Λ derivation.** Test whether Ω_Λ can arise from a phase-sector ratio, renormalized curvature residue, or observable-domain mapping without circular input.
3. **Null-domain likelihood.** Map D_γ to D_C, D_M, D_L, D_A and fit Pantheon+ and DESI DR2.
4. **CMC phase calculation.** Compute $\Theta(t)$ for a concrete compact S^3 TT bundle or density matrix and verify Eq. (40).
5. **Closed-mode Boltzmann implementation.** Implement S^3 mode sums in CLASS/CAMB-like pipelines and compare low-ℓ spectra to Planck likelihoods.
6. **Model comparison.** Run Bayesian comparison against ΛCDM for parameters (ε, p, R_c) with conservative priors, including $\varepsilon > 0$ and $p > 0$ for the proxy tested here.

A Notation Dictionary

Symbol	Meaning	First defined
φ	golden ratio $(1+\sqrt{5})/2$	Sec. 3
$R_n = \ell_{\rm P}\varphi^n$	φ-hierarchy radius	Sec. 3
n_R	geometric level $\ln(R/\ell_{\rm P})/\ln\varphi$	Sec. 3
$n_A = n_R - 207$	amplitude-level hypothesis	Sec. 3
$C_{\rm EH} = 3/(8\pi)$	spatial EH curvature coefficient	Sec. 5
$R_{\rm eff} = R_H/\sqrt{\Omega_\Lambda}$	effective curvature-energy radius	Sec. 6
R_c	global FLRW curvature radius	Sec. 7
$\chi_H = \sqrt{\Omega_\Lambda}$	Hubble angle on auxiliary $S^3(R_{\rm eff})$	Sec. 7
$K(\chi)$	S^3 volume-growth kernel	Sec. 2
D_γ	light-front arc-length variable	Sec. 9
$F_{\rm dyn}(a)$	phenomenological DE evolution factor	Sec. 11

B Independent Numerical Check

The central identities can be reproduced from

$$\ell_{\rm P} = \sqrt{\frac{\hbar G}{c^3}}, \qquad \rho_{\rm P} = \frac{c^7}{\hbar G^2}, \qquad \rho_{\rm P}\ell_{\rm P}^2 = \frac{c^4}{G}, \tag{51}$$

$$\rho_{\rm crit} = \frac{3H_0^2c^2}{8\pi G}, \qquad \rho_{\Lambda,\rm obs} = \Omega_\Lambda\rho_{\rm crit}, \qquad R_{\rm eff} = \sqrt{\frac{C_{\rm EH}c^4}{G\rho_{\Lambda,\rm obs}}}. \tag{52}$$

With the values of Table 3,

$$\frac{\rho^{(3)}_{\rm EH}(R_{\rm eff})}{\rho_{\Lambda,\rm obs}} = 1.0000000000 \tag{53}$$

within rounding precision.

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